Chapter 12, Problem 47P

Problem

The following three parallel-connected three-phase loads are fed by a balanced three-phase source:

Load 1: 250 kVA, 0.8 pf lagging
Load 2: 300 kVA, 0.95 pf leading
Load 3: 450 kVA, unity pf

If the line voltage is 13.8 kV, calculate the line current and the power factor of the source. Assume that the line impedance is zero.

Step-by-step solution

Step 1 of 1

$$Pf = 0.8\ [Lagging] \Rightarrow \theta = Cos^{-1}(0.8) = 36.87°$$

$$S_1 = 250\angle 36.87° = 200 + j150\ KVA\backslash$$

$$Pf = 0.95\ [Leading] \Rightarrow \theta = Cos^{-1}(0.95) = -18.19°$$

$$S_2 = 300\angle -18.9° = 285 - j93.65\ KVA$$

$$Pf = 1.0 \Rightarrow \theta = Cos^{-1}(1) = 0°$$

$$S_3 = 450\ KVA$$

$$S_T = S_1 + S_2 + S_3 = 935 + j56.35$$

$$= 936.7\angle 3.45\ KVA$$

$$|S_T| = \sqrt{3} V_L I_L$$

$$I_L = \frac{936.7\times 10^3}{\sqrt{3}(13.8\times 10^3)} = 39.19\ Arms$$

$$Pf = Cos\theta = Cos(3.45°) = 0.9982\ [Lagging]\ .$$